

Full Stack Development with MERN

Project Documentation

1. Introduction

• **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

• **Team Members:**

- Tarangini PR - Team Lead & ServiceNow System Administrator (Overall project orchestration, instance setup, and deployment verification)
- Charunethra T – ServiceNow Process Architect (Workflow logic design, Flow Designer trigger & task action setup)
- Prasanthi P – Service Catalog Specialist (Catalog item configuration, Process Engine linking in Maintain Items)
- Varshini M – Quality Assurance & Testing Lead (End-to-end user request testing, approval routing validation, and SCTASK assignment verification)
- Laya Abraham – Technical Documentation & UI Specialist (Architecture diagramming, technical document creation, and demo video production)

2. Project Overview

• **Purpose:** The current IT procurement process lacks automation, causing delays and manual overhead when processing standard laptop requests. This project implements an automated workflow within ServiceNow using Flow Designer to instantly create, configure, and assign catalog fulfillment tasks (SCTASK) to the Hardware team upon request approval.

Features:

- Automated trigger execution on Service Catalog submissions.
- Auto-generation of Catalog Tasks (SCTASK) with short description set to "Laptop needs to Configured".
- Direct task routing and assignment to the Hardware group.
- Automatic state transition updates linking Request (REQ), Requested Item (RITM), and Catalog Task (SCTASK).
- Seamless integration with Service Portal and Employee Center catalogs.

3. Architecture

- **Frontend:** ServiceNow Service Portal and Employee Center UI components providing the catalog interface for end users to browse hardware items and submit orders.

- **Backend:** ServiceNow Platform Process Engine and Flow Designer (Standard Laptop task flow running as System User) executing backend workflow automation.
- **Database:** ServiceNow relational relational record schema mapping parent-child relationships across the Request (sc_request), Requested Item (sc_req_item), and Catalog Task (sc_task) database tables.

4. Setup Instructions

- **Prerequisites:** ServiceNow Developer Instance (Zurich/Australia releases or above).
System Administrator (admin) role privileges.
- **Installation:**
- Log into your ServiceNow Developer Instance.
- Navigate to Process Automation --> Flow Designer.
- Create a new flow titled Standard Laptop task in the Global scope with Run As set to System User.
- Set Trigger to Service Catalog.
- Add Action Create Catalog Task, drag/drop the Requested Item Record, and set the fields:
 - Short Description: Laptop needs to Configured
 - Description: Laptop needs to Configured
 - Assignment Group: Hardware
 - Approval: Approved
- Save and Activate the flow.
- Navigate to Maintain Items, open Standard Laptop, go to the Process Engine tab, and assign the flow Standard Laptop task.

5. Folder Structure

Client (Presentation Layer):

- Service Catalog / Hardware – Catalog view and item ordering forms.
- Service Portal (SP) / Employee Center (EC) – User-facing order entry portals.

Server (Business Logic Layer):

- Flow Designer / Flows / Standard Laptop task – Main automation flow logic.
- Data Schema / Tables – sc_request, sc_req_item, sc_task records.

6. Running the Application

- Triggering the Workflow: Navigate to Service Catalog --> Hardware --> Standard Laptop and click Order Now.
- Approving Request: Open the generated Request record (REQ), scroll to the Approvers section, right-click the record, and click Approve.
- Verifying Output: Open the child Requested Item (RITM) record, scroll to Catalog Tasks, and verify the automatically generated SCTASK.

7. API Documentation

- Trigger Event: Service Catalog Submission Event
- Internal Table REST/Data APIs:
 - POST /api/now/table/sc_request – Standard Catalog Request generation.
 - POST /api/now/table/sc_req_item – Requested Item record instantiation.
 - POST /api/now/table/sc_task – Automated Catalog Task generation via Flow Designer.
- Payload Fields:
 - short_description: "Laptop needs to Configured"
 - assignment_group: "Hardware"
 - approval: "Approved"

8. Authentication

- User Authentication: Handled natively via ServiceNow Role-Based Access Control (RBAC) and system login sessions.
- Execution Authorization: The Standard Laptop task flow is configured to Run As: System User to ensure task generation executes elevated automation privileges regardless of the end user's system role.

9. User Interface

Service Catalog order submission page for Standard Laptop showing the "Order Now" button.

The screenshot shows the ServiceNow Service Catalog interface. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The main content area displays the 'Lenovo - Carbon x1' product page. The product description states: 'The x1 Carbon is Lenovo's lightest ThinkPad yet. It provides a QHD display that fights glare and weighs less than three pounds. Ideal for most computing tasks, and highly mobile.' Technical specifications listed include: Intel core i5 processor, 512GB solid state drive (SSD), and Backlit keyboard. Under 'Optional Software', there are checkboxes for 'Adobe Acrobat' and 'Adobe Photoshop'. A section for 'Additional software requirements' is also present. On the right, the 'Order this Item' sidebar shows the price as '\$1,100.00 + \$100.00 Annually', a quantity of '1', a subtotal of '\$1,100.00 + \$100.00 Annually', and a delivery time of '0 Days'. The 'Order Now' button is highlighted in blue, and there is an 'Add to Cart' button below it. The 'Shopping Cart' section at the bottom right is currently empty.

Order placement confirmation screen displaying Request Number REQ0010001 and estimated delivery details.

servicenow All Favorites History Admin : Order Status: REQ0010001 Search

Order Status [Back to Catalog](#) [Continue Shopping](#) [Home](#)

Thank you, your request has been submitted

Order Placed: 2026-10-01 00:21:03
Request Number: [REQ0010001](#) ☆
Estimated Delivery Date of Complete Order: 2026-10-01

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
Lenovo - Carbon x1	2026-10-01		\$1,100.00 +\$100.00 Annually	1	\$1,100.00 +\$100.00 Annually
Total					\$1,100.00 +\$100.00 Annually

[Back to Catalog](#) [Continue Shopping](#) [Home](#)

Service Request record view showing state set to Pending Approval.

servicenow All Favorites History Admin : Order Status: REQ0010001 Search

Request REQ0010001 [Follow](#) [Update](#) [Add New Item](#) [Cancel Request](#) [Copy](#) [Delete](#)

Number: REQ0010001
Requested for: System Administrator
Location:
Due date: 2026-10-01 00:21:02
Price: \$1,100.00
Description:
Short description:
Special instructions:
Work notes: Work notes

Opened: 2026-10-01 00:21:03
Opened by: System Administrator
Approval: Requested
Request state: Pending Approval

Post

Request record (REQ) showing the Approvers tab and manager approval action.

servicenow All Favorites History Admin : Request - REQ0010001 Search

Request REQ0010001 [Discuss](#) [Follow](#) [Update](#) [Add New Item](#) [Cancel Request](#) [Copy](#) [Delete](#)

Impact: 3 - Low
Opened by: System Administrator
Priority: 4 - Low
State: Open

Update Add New Item Cancel Request Copy Delete

Requested Items (1) Approvals (1) Recurring Prices (1)

State:
Approval for = REQ0010001
Show Matching
Filter Out
Copy URL to Clipboard
Copy sys_id
Assign Tag
Approve
Reject

Approver	Comments	Created
Eric Schroeder		2026-10-01 00:21:08

1 to 1 of 1

10. Testing

Strategy: End-to-end integration testing in a ServiceNow personal developer instance.

Test Cases:

- Order Submission: Validate that ordering a standard laptop generates a valid REQ and RITM record.
- Approval Trigger: Validate that approving the request triggers the Standard Laptop task flow.
- Field Population: Verify that SCTASK short description equals "Laptop needs to Configured" and assignment group equals "Hardware".

11. Screenshots or Demo

Creating a new workflow titled "Standard laptop task" in Workflow Studio.

The screenshot shows the 'New Flow' configuration screen in ServiceNow Workflow Studio. The left sidebar contains a 'Return home' button and a visual flow diagram with two steps. The main area is titled 'Let's get the details for your flow' and contains the following fields:

- Flow name:** Standard laptop task
- Application:** Global
- Description:** Describe your flow.
- Hide additional properties:** A link to expand more options.
- Protection:** -- None --
- Run as:** System user

At the bottom right, there are 'Cancel' and 'Build flow' buttons. An 'Activate Windows' watermark is visible in the bottom right corner.

Configuring the "Create Catalog Task" action with Short Description, Assignment group (Hardware), and Approval parameters.

The screenshot shows the configuration screen for the 'Create Catalog Task' action within the 'Standard laptop task' workflow. The left sidebar shows the workflow steps, with the first step being 'Create Catalog Task'. The main area is titled 'Action Properties' and contains the following fields:

- Action:** Create Catalog Task
- Table Name:** Catalog Task [sc_task]
- Requested Item (Requested Item):** Trigger -> Requested Item ->
- Short Description:** Laptop need to Configured
- Description:** Laptop need to Configured
- Assignment group:** Hardware
- Approval:** Approved
- Wait:** ☒

At the bottom left, there is a '+ Add field value' button. The bottom status bar shows 'Status: Draft' and 'Application: Global'. On the right, there is a 'Data' panel with a 'Collapse All' button and a list of variables. An 'Activate Windows' watermark is visible in the bottom right corner.

Linking the active "Standard laptop task" flow to the Catalog Item under the Process Engine tab in Maintain Items.

servicenow All Favorites History Admin Catalog Item - Standard Laptop Search

Catalog Item Standard Laptop Copy Try It Update Edit in Catalog Builder Delete

Build and modify items faster with the improved [Catalog Builder](#).

Name Standard Laptop Application Global

Catalogs Service Catalog Active ☒

Category Hardware Roles

State -- None -- Fulfillment automation level Unspecified

Checked out -- None --

Owner

Secondary owners

Item Details **Process Engine** Picture Pricing Portal Settings

Flow Standard laptop task

Workflow

Requested Item record (RITM0010001) showing the generated Catalog Task under the Catalog Tasks tab.

servicenow All Favorites History Admin Requested Item - RITM0010001 Search

Requested Item RITM0010001 Discuss Follow Update Delete

Impact 3 - Low
Opened by System Administrator
State Open

Update Delete

Related Links
[Flow Context](#)

Catalog Tasks (1) Approvers

Number Search Actions on selected rows... New

Request Item = RITM0010001

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010001	Hardware	(empty)	Laptop need to Configured	2026-10-01 00:25:00	(empty)	

1 to 1 of 1

Auto-generated Catalog Task record (SCTASK0010001) assigned to the Hardware group for fulfillment.

servicenow All Favorites History Admin : Catalog Task - SCTASK0010001 Search

Catalog Task
SCTASK0010001

Number: SCTASK0010001

Assigned to: [Search]

Configuration item: [Search]

Active: ☒

Approval: Approved

Priority: -- None --

State: Open

Request item: RITM0010001 [Search]

Requested for: System Administrator [Search]

Short description: Laptop need to Configured

Description: Laptop need to Configured

Work notes: Work notes

☐ Comments (Customer visible) **Post**

Activities: 1 System

Field changes • 2026-10-01 00:25:00

12. Known Issues

- Manual right-click approval in demo instances requires Administrator session overriding.
- Flow execution delays of 1–2 seconds may occur if background job workers are heavily loaded.

13. Future Enhancements

- Automated email notifications sent to the Hardware group upon task creation
- Integration with asset management tables to auto-reserve available laptop serial numbers.
- Adding SLA workflow timers to track hardware configuration turn-around times.